

Jayam B. Patel

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EDUCATION

University of Massachusetts Lowell

Lowell, MA

Bachelor of Science in Computer Science, GPA: 3.98/4.00

Expected May 2028

- **Coursework:** Computing I–IV, Discrete Structures I–II, Probability & Statistics, Computer Security, Logic Design, Foundations of Computer Science
- **Honors:** Chancellor's List, Billerica Dollars for Scholars, Demoulas Market Basket Scholarship

LANGUAGES & TECHNOLOGIES

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash, C/C++, HTML/CSS, YAML/Jinja2

AI & ML: Claude Code, LiteLLM, AWS Bedrock, MCP Servers, Multi-Agent Systems, OpenAI / Claude / Gemini APIs

Data & Observability: PostgreSQL, MySQL, Patroni HA, JSONB, ETL Pipelines, Grafana, Prometheus, Data Analytics

Cloud & DevOps: AWS (EC2, EBS, SSM, IAM), Ansible, Semaphore, Docker, Docker Swarm, Ceph, GitLab CI/CD, Git/GitHub

Security & Practices: OpenBao, Keycloak / OIDC, Headscale / Tailscale, JIT Credentials, SDLC, Domain-Driven Design

EXPERIENCE

Software Engineer, Co-op

July 2026 – December 2026

Analogic Corporation

Salem, NH

- Built and deployed a self-hosted Headscale/Tailscale VPN gateway (DERP relay on port 443, scoped subnet router) that gives the team private network access into a cloud environment, with OIDC single sign-on and vault-managed secrets (OpenBao); debugged a chain of stacked container deploy failures to reach a stable running state.
- Authored the runtime that drives an internal AI-agent engagement through its full lifecycle (a git-backed lock ceremony, automated merge-request creation, and clean teardown), and designed a just-in-time credential flow that mints, tightly scopes, and revokes short-lived git tokens per engagement.
- Shipped self-service cloud provisioning on an internal platform: Semaphore/Ansible templates and a multi-stage personal-instance workflow (launch, pause/resume, snapshot-first teardown), and migrated the tool's configuration to version-controlled config-as-code with a SQLite-to-PostgreSQL datastore cutover.
- Authored 8 Ansible roles across provisioning, cleanup, routing, and ingress, including race-safe, idempotent automation (AWS SSM) that publishes network routes and reaps orphaned resources across 6 AWS resource classes, reconciling against live infrastructure state instead of stored outputs.
- Built 2 Grafana/Prometheus dashboards end-to-end for LLM (LiteLLM) spend and Docker Swarm health, tracking spend across 6 model routes, and repaired the underlying metrics pipeline (counter names, rate intervals, per-container scraping) so panels reported real data.
- Eliminated a class of command-injection risk by routing CI-triggered automation through a shared file-ingress gateway, and contributed reliability fixes across team-owned systems including a high-availability PostgreSQL (Patroni) failover fix, a coordinated cross-codebase identifier rename with dual-tagging, and CI/CD test-gate hardening.

AI Specialist

June 2025 – December 2025

Exponentia Strategies

Boston, MA (Remote)

- Built marketing-automation pipelines that route content generation across multiple LLM APIs, producing LinkedIn content and personalized 30-day content strategies from user inputs.
- Engineered n8n workflows linking a web frontend to backend LLM processing, with user authentication, session validation, and multi-stage generation.
- Designed MySQL schemas for user profiles and content plans, and built full-stack PHP/JavaScript/AJAX applications serving real-time generated content.

Creative AI Web Builder

May 2025 – December 2025

Luxeciaga

Hartford, WI (Remote)

- Built and deployed responsive, mobile-first websites in custom HTML/CSS/JavaScript, tuning performance and behavior across browsers.
- Implemented SEO strategies and Google Analytics tracking to drive data-informed design and performance improvements.

PROJECTS

Smart Degree | Python, PostgreSQL, Supabase, JSONB, Vercel, Data Engineering

Winter 2025

- Built a transcript-analytics system extracting structured academic data from unstructured PDFs, normalizing 2,000+ university courses into a PostgreSQL/JSONB database, computing GPA trends and degree-completion metrics, deployed full-stack on Vercel with a Supabase backend.